Theoretical Neuroscience : Chapter 1

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Objective : Basic statistics describing how stimulus evoke spike sequences.

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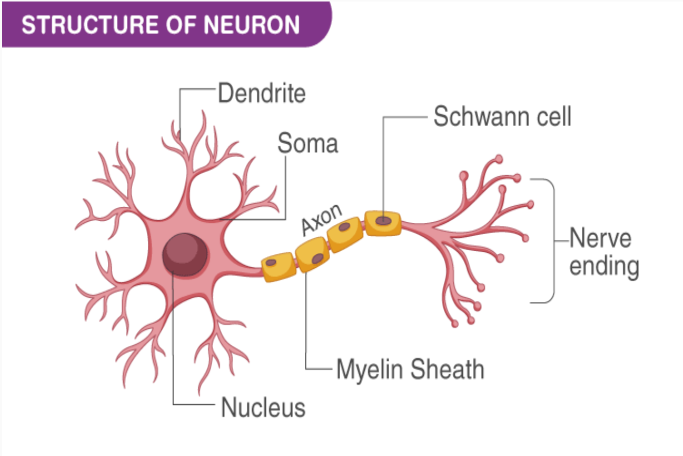
Structure of a Neuron

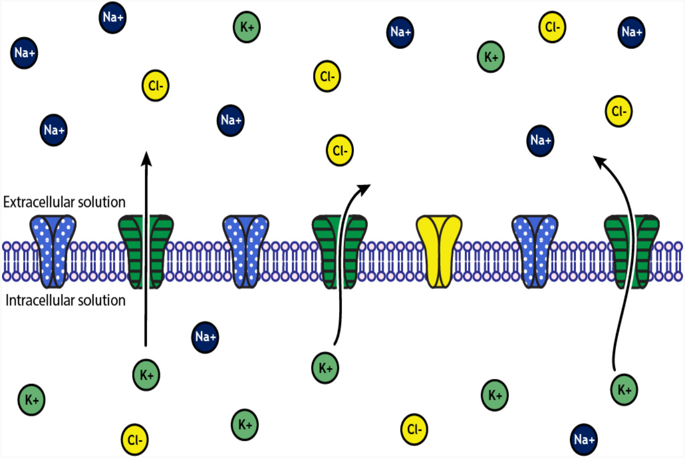


From https://byjus.com/biology/neurons/

Axons and dendrites

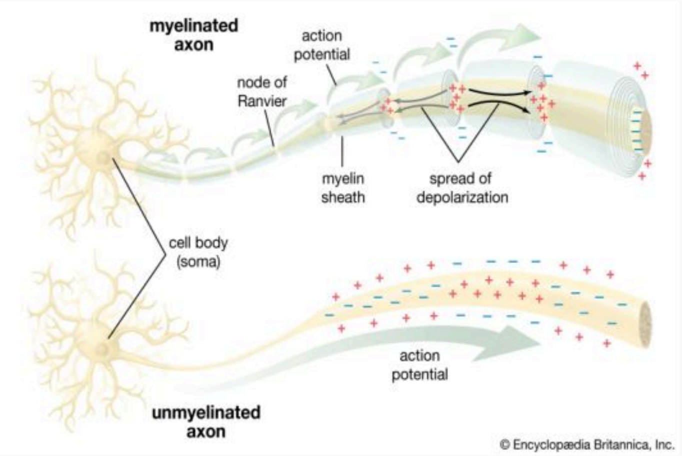
- One axon per one neuron vs many dendrites per one neuron.
- Axon is long vs dendrites are short.
- Dendrites receive messages from other cells.
- Axons pass messages from the cell body to other neurons or muscles.
- Myelin Sheath covers the axon and it helps speed the neural impulses.

Action Potential



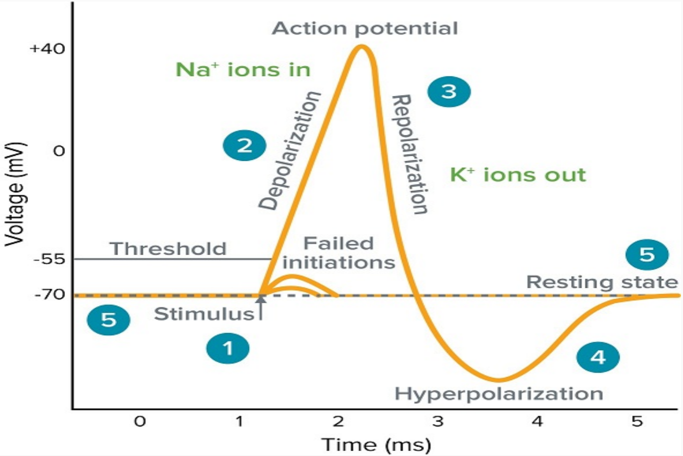
From https://www.britannica.com/science/action-potential

Action Potential



From https://www.britannica.com/science/action-potential

Action Potential



From https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential

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Spikes

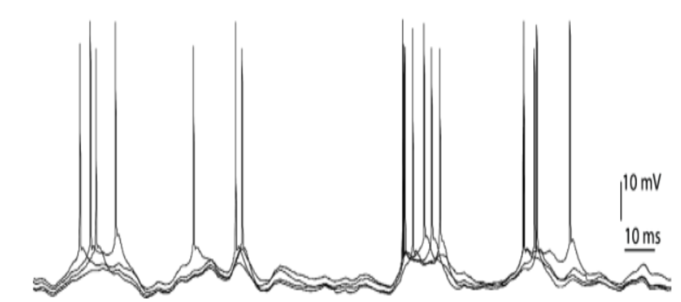
Because of the action potential, there arise a sequence of spikes represented by the dirac delta functions, which is called a neural response function

$$\rho(t) = \sum_{i=1}^{n} \delta(t - t_i).$$

For any well-behaved function h, we will use the identity

$$\sum_{i=1}^{n} h(t - t_i) = \int_{-\infty}^{\infty} d\tau h(\tau)\rho(t - \tau).$$

Spikes



From https://neuronaldynamics.epfl.ch/online/Ch7.S1.html
Repetition by four stimuli

Spikes

We introduce several terminologies. Keep in mind that

$$\text{Stimulus} \to \text{Firing} \to \text{Spike}$$

- Spike-count rate r.

$$r = \frac{n}{T} = \frac{1}{T}\int_0^T d\tau\ \rho(\tau).$$

Thus, the the spike-count rate is the time average of the neural response function over the duration of the trial.

- Averaging over the trials that use the same stimulus

$$\langle \rho(t) \rangle.$$

- Average firing rate $\langle r \rangle$

$$\langle r \rangle = \frac{\langle n \rangle}{T} = \frac{1}{T}\int_0^T d\tau \langle \rho(\tau) \rangle = \frac{1}{T}\int_0^T dt\ r(t).$$

- Firing rate $r(t)$

$$r(t) = \frac{1}{\Delta t} \int_{t}^{t+\Delta t} d\tau \langle \rho(\tau) \rangle.$$

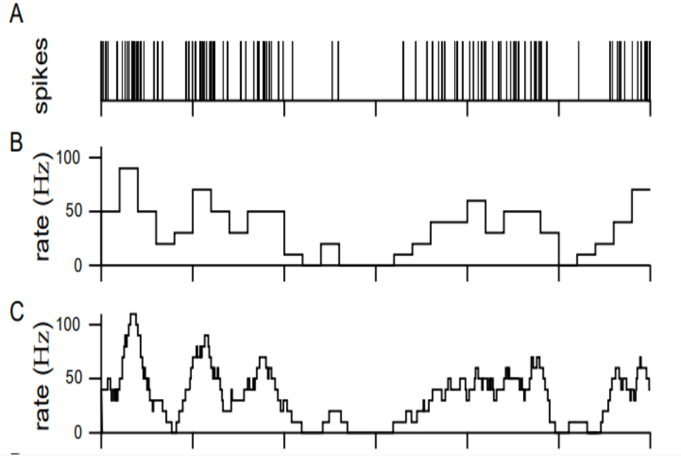
Observe that in the limit $\Delta t \to 0$, we have $r(t) = \langle \rho(\tau) \rangle$. However, Δt should be sufficiently large so that there are sufficient numbers of spikes within the interval. However, we will use that for a good enough function h, we have

$$\int d\tau h(\tau) \langle \rho(t-\tau) \rangle = \int d\tau h(\tau) r(t-\tau).$$

- $r(t)\Delta t$ is the probability that a spike occurs during the time interval whose length is Δt.

Measuring Firing Rates

- How to measure the firing rate $r(t)$?



Measuring Firing Rates

- B. Divide the time interval with equal size, and count.
- Or, we approximate $r(t)$ by using the kernel as follows:

$$r_a(t) = \int_{-\infty}^{\infty} d\tau \omega(\tau)\rho(t-\tau),$$

which is called a linear filter with the filter kernel ω.

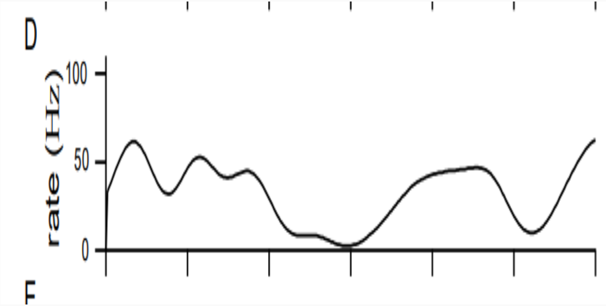
- C. Take the window function as

$$\omega(t) = \begin{cases} 1/\Delta t & \text{if} \quad -\Delta t/2 \leq t < \Delta t/2 \\ 0 & \text{otherwise.} \end{cases}$$

• Or, we may take

$$\omega(\tau) = \frac{1}{\sqrt{2\pi}\sigma_\omega} \exp\left(- \frac{\tau^2}{2\sigma_\omega^2} \right),$$

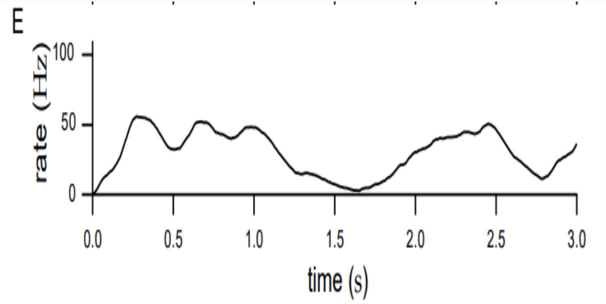
which gives the approximation



- Or, if we only take the past effect, we take

$$\omega(\tau) = [\alpha^2 \tau \exp(-\alpha\tau)]_+,$$

which gives the approximation



Tuning Curves

- The neural response tuning curve:

The average firing rate written as a function of s, $\langle r \rangle = f(s)$.

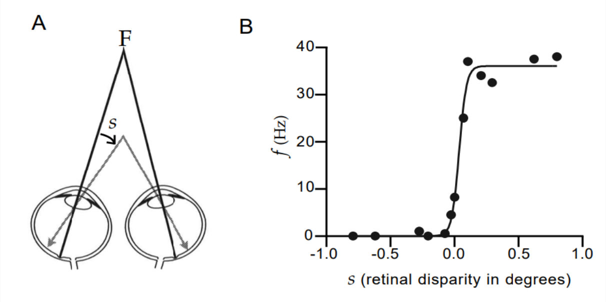
- For example,

$$f(s) = r_{max} \exp\left(-\frac{1}{2}\left(\frac{s - s_{max}}{\sigma_f}\right)^2 \right),$$

- We assume that the stimulus function s is periodic in period T.

Tuning Curves

- Consider the retinal disparity,



Here, we take

$$f(s) = \frac{r_{\max}}{1 + \exp((s_{1/2} - s)/\Delta_s)}.$$

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• Spike-triggered average stimulus, $C(\tau)$, is the average value of the stimulus a time interval τ before a spike is fired.

$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^{n} s(t_i - \tau) \right\rangle \approx \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^{n} s(t_i - \tau) \right\rangle,$$

where we assumed $n \approx \langle n \rangle$. Observe that

$$\begin{aligned} C(\tau) &= \frac{1}{\langle n \rangle} \int_0^T dt \langle \rho(t) \rangle s(t - \tau) \\ &= \frac{1}{\langle n \rangle} \int_0^T dt r(t) s(t - \tau). \end{aligned}$$

- The correlation function of the firing rate and the stimulus is

$$Q_{rs}(\tau) = \frac{1}{T}\int_0^T dt\ r(t)s(t+\tau).$$

Now, using that $\langle r\rangle = \langle n\rangle / T$, we see that

$$C(\tau) = \frac{1}{\langle r\rangle} Q_{rs}(-\tau),$$

therefore, the spike-triggered average stimulus $C(\tau)$ is essentially the reverse correlation function.

White-noise stimuli

- The white-noise stimulus has the defining characteristic that its value at any one time is uncorrelated with its value at any other time. Define the stimulus autocorrelation function

$$Q_{ss}(\tau) = \frac{1}{T}\int_0^T dt\ s(t)s(t+\tau).$$

For the white noise, the autocorrelation function is 0 in the range $-T/2 < \tau < T/2$ except when $\tau = 0$, i.e.

$$Q_{ss}(\tau) = \sigma_s^2 \delta(\tau).$$

This is equivalent to saying that white nose has equal power at all frequencies.

The Power spectrum of White noise

The Fourier transform of the stimulus autocorrelation function Q_{ss} is

$$\hat{Q}_{ss}(\omega) = \frac{1}{T}\int_{-T/2}^{T/2} d\tau\, Q_{ss}(\tau)\exp(\mathrm{i}\omega\tau),$$

which is called the power spectrum. For the white noise $Q_{ss}(\tau) = \sigma_s^2\delta(\tau)$,

$$\hat{Q}_{ss}(\omega) = \frac{\sigma_s^2}{T}\int_{-T/2}^{T/2} d\tau\, \delta(\tau)\exp(\mathrm{i}\omega\tau) = \frac{\sigma_s^2}{T}.$$

The Power spectrum of White noise

In general, consider the Fourier transform of the stimulus

$$\hat{s}(\omega) = \frac{1}{T}\int_0^T d\tau\ s(t)\exp(\mathrm{i}\omega t).$$

Then, we compute that

$$\begin{aligned}\hat{Q}_{ss}(\omega) &= \frac{1}{T}\int_0^T dt\ s(t)\frac{1}{T}\int_{-T/2}^{T/2} d\tau\ s(t+\tau)\exp(\mathrm{i}\omega\tau)\\ &= \frac{1}{T}\int_0^T dt\ s(t)\exp(-\mathrm{i}\omega t)\frac{1}{T}\int_{-T/2}^{T/2} d\tau\ s(t+\tau)\exp(\mathrm{i}\omega(t+\tau))\\ &= \bar{\hat{s}}(\omega)\hat{s}(\omega) = |\hat{s}(\omega)|^2.\end{aligned}$$

In practice, we insstead estimate $\hat{Q}_{ss}(\omega)$ by

$$\frac{\Delta T}{T}\left|\sum_{m=1}^{T/\Delta t} s(t)\exp(-\mathrm{i}\omega m\Delta t)\right|^2.$$

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Spike-Train Statistics

• We want to know the probabilities corresponding to every sequence of spikes that can be evoked by the stimulus.

• The probability that a sequence of n spikes occurs with spike i falling between times t_i and $t_i + \Delta t$ for each $i = 1, 2, \cdots, n$ is given by the density as

$$P[t_1, \cdots, t_n] = p[t_1, \cdots, t_n](\Delta t)^n.$$

• We consider the Poisson process and separate two cases depending on whether the firing rate $r(t)$ is constant over time or not. (Homogeneous/Inhomogeneous).

Homogeneous Poisson process

- In this case $r(t) = r$.
- Then, the probability $P[t_1, \cdots, t_n]$ is independent of the explicit numbers t_i. Thus, if

$$0 \leq t_1 \leq t_2 \leq \cdots \leq t_n \leq T,$$

then, we have the relationship

$$P[t_1, \cdots, t_n] = n! P_T[n] \left(\frac{\Delta t}{T} \right)^n,$$

where $P_T[n]$ is the probability that an arbitrary sequence of exactly n spikes occurs within a trial of duration T.
- Indeed, we can deduce that

$$P_T[n] = \frac{(rT)^n}{n!} \exp(-rT).$$

Derivation of $P_T[n]$

1. Divide the time T into M bins of size $\Delta t = T/M$. Assume that Δt is small enough so that we never get two spikes within any one bin.
2. Note that $P_T[n]$ is the product of : the probability of generating n spikes within a specified set of the M bins, the probability of not generating spikes in the remaining $M - n$ bins, and a combinatorial factor equal to the number of ways of putting n spikes into M bins.
3. Thus,

$$P_T[n] = \lim_{\Delta t \to 0} \binom{M}{n} (r\Delta t)^n (1 - r\Delta t)^{M-n}$$

4. In $\Delta t \to 0$ with $M\Delta t = T$, we approximate $M - n \approx M = T/\Delta t$. Then,

$$\lim_{\Delta t \to 0} (1 - r\Delta t)^{M-n} = \exp(-rT).$$

Homogeneous Poisson process

- Using $P_T[n]$, we compute the variance of the spike count for spikes counted over an interval of duration T:

$$\sigma_n^2 = \langle n^2 \rangle - \langle n \rangle^2 = rT.$$

Sketch of proof) We note that

$$\langle n \rangle = \sum_{n=0}^{\infty} n P_T[n] = \sum_{n=0}^{\infty} \frac{n(rT)^n}{n!} \exp(-rT),$$

and also,

$$\sigma_n^2(T) = \sum_{n=0}^{\infty} n^2 P_T[n] - \langle n \rangle^2.$$

Computing moments

Using the moment generating function

$$g(\alpha) := \sum_{n=0}^{\infty} \frac{(rT)^n \exp(\alpha n)}{n!} \exp(-rT).$$

We observe that

$$g(\alpha) = \exp(-rT)\exp(rTe^{\alpha}).$$

Thus,

$$\frac{dg}{d\alpha} = rT,$$
$$\frac{d^2 g}{d\alpha^2} = (rT)^2 + rT.$$

Inhomogeneous Poisson Statistics

- When $r(t)$ depends on time, different sequences of n spikes occur with different probabilties and $p[t_1, \cdots, t_n]$ depends on the spike times.

$$p[t_1, \cdots, t_n] = \exp\left(- \int_0^T dt r(t) \right) \prod_{i=1}^{n} r(t_i).$$

Sketch of derivation) First, we compute the probability that no spikes are generated during the time interval from t_i to t_{i+1}. Dividie the time interval by $M\Delta t = t_{i+1} - t_i$. Then observe that

$$P[\text{no spikes}] = \prod_{m=1}^{M} (1 - r(t_i + m\Delta t)\Delta t).$$

Take logarithm and use $\ln(1-x) \approx -x$. Then as $\Delta t \to 0$.

$$\ln P[\text{no spikes}] = -\sum_{m=1}^{M} r(t_i + m\Delta t)\Delta t \to -\int_{t_i}^{t_{i+1}} dt r(t).$$

Inhomogeneous Poisson Statistics

Finally, we have the probability density

$$p[t_1, \cdots, t_n] = \exp\Big(-\int_0^{t_1} dt r(t)\Big)\exp\Big(-\int_{t_n}^{T} dt r(t)\Big) r(t_n)$$
$$\times \prod_{i=1}^{n-1} r(t_i)\exp\Big(-\int_{t_i}^{t_{i+1}} dt r(t)\Big)$$
$$= \exp\Big(-\int_0^{T} dt r(t)\Big)\prod_{i=1}^{n} r(t_i).$$

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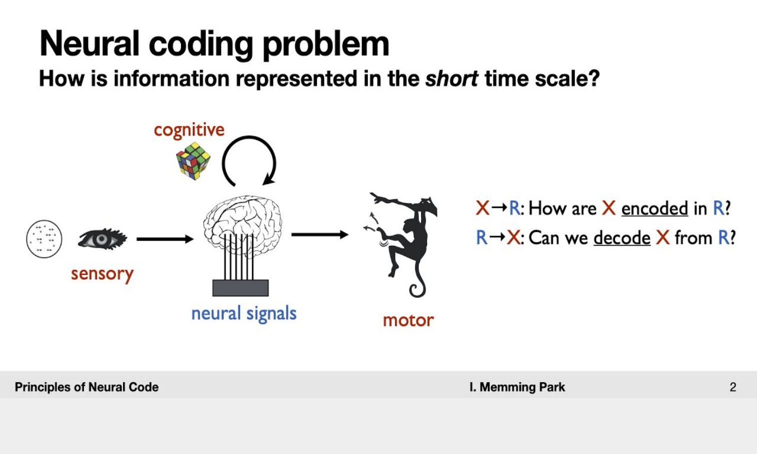
Neural Coding



From https://www.youtube.com/watch?v=DIFxUEdGImQ

Neural Coding

- The nature of the neural code is a topic of intense debate within the neuroscience community.
- Whether neurons use rate coding vs tempopral coding.
- That is, individual action potentials and neurons encode independently of each other vs correlations between different spikes and neurons carry significant amounts of information.
- However, both happens in reality.

Neural Coding

- Independent-spike code vs Correlation code

- The neural response and its relation to the stimulus are completely characterized by the probability disstribution of spike times as a function of the stimulus.

- If spike generation is described by an inhomogeneous Poisson process, then

$$p[t_1, \cdots, t_n] = \exp\left(-\int_0^T dt r(t)\right) \prod_{i=1}^{n} r(t_i),$$

where $r(t)$ is the time-dependent firing rate.

Independent-spike code vs Correlation code

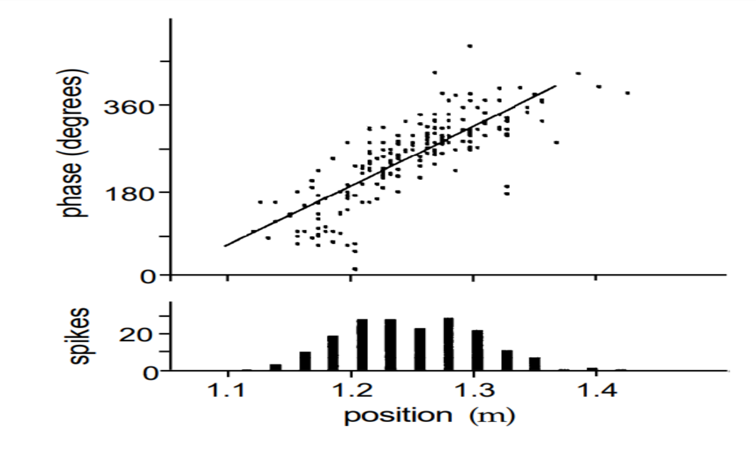
• In this case, $r(t)$ contains all the information about the stimulus that can be extracted from the spike train.
$\Rightarrow$ Independent-spike code.

• If individual spikes do not encode independently of each other, we call the neural code
$\Rightarrow$ Correlation code.

• Independent-spike coding is regarded as quite convincing at least as a fairly accurate approximation.

• However, this does not mean that we do not consider the ensemble of neurons.

Neural Coding



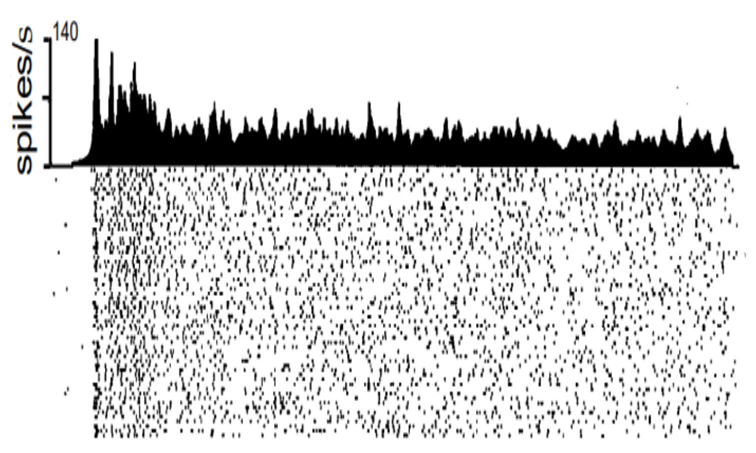
Synchronous firing in the rat hippocampus

However, synchronization does not mean spikes are correlated.

Temporal Codes

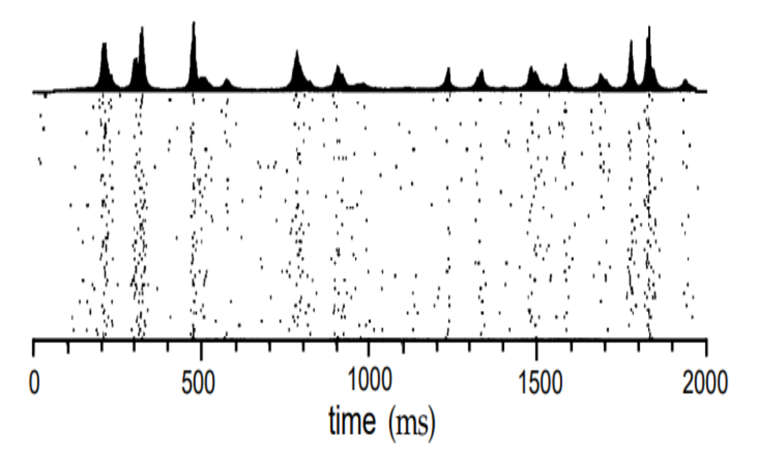
- When precise spike timing or high-frequency firing-rate fluctuations are found to carry information, the neural code is often identified as a temporal code.
- The temporal coding is distinct and independent from the issue of independent-spike coding.
- We distinguish temporal coding from rate coding according to whether the firing rate $r(t)$ changes slowly or rapidly.

rate coding vs temporal coding



When $r(t)$ varies slowly, rate code.

rate coding vs temporal coding



When $r(t)$ varies rapidly, temporal code.

Issues in Neural Coding

- Q. how to distinguish rate from temporal?

1. Use spikes to distinguish slow from rapid.
   Each peak corresponds to the firing of only one action potential.
   Spike frequency = Peak frequency.

2. Use stimulus.
   Spike timing scale < fastest time for variations of the stimulus.

- The central challenge is to identify relationships between the firing patterns of different neurons in a responding population and to understand their significance for neural coding.

Thank you for the listening!